King Fahd University of Petroleum & Minerals
Department of Mathematics and Statistics

Math 101
Final Exam
Term 102
Thursday, June 9, 2011
Net Time Allowed: 180 minutes

MASTER VERSION

1. If $x^2 + 1 \leq f(x) - 2x \leq 3x^4 - 1$ for all $x \in (-\infty, \infty)$, then $\lim_{x \to 1} f(x) =$

   (a) 4

   (b) -3

   (c) 1

   (d) 0

   (e) Does not exist

2. The function $g(x) = \dfrac{\sqrt{x^2 - 4}}{x^2 - 3x}$ is **continuous** on

   (a) $(-\infty, -2] \cup [2, 3) \cup (3, \infty)$

   (b) $(-\infty, \infty)$

   (c) $(-\infty, 0) \cup (0, 3) \cup (3, \infty)$

   (d) $(-\infty, -2] \cup [2, \infty)$

   (e) $[-2, 0) \cup (0, 2]$

3. $\lim_{x\to 0^-} \tan^{-1}(e^{1/x}) =$

(a) 0

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) $-\frac{\pi}{2}$

(e) ∞

4. The equation of the **tangent line** to the curve $y = x^4 - 2x$ at $x = 1$ is

(a) $y = 2x - 3$

(b) $y = 3x - 4$

(c) $y = x - 2$

(d) $y = -4x + 3$

(e) $y = -2x + 1$

5. If $f(x) = \begin{cases} \cos x & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0, \end{cases}$ then $f'(0) =$

(a) Does not exist

(b) 0

(c) 2

(d) -1

(e) 3

6. If $f(x) = \dfrac{1}{2+e^x}$, then $\lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h} =$

(a) $\dfrac{-e^x}{(2+e^x)^2}$

(b) $\dfrac{1}{3}$

(c) $\dfrac{-2}{(2+e^x)^2}$

(d) $2+e^x$

(e) e^{-x}

7. Let $y = Ax^2 + Bx + C$. If

$$y'' + y' - 2y = x^2$$

then $A + B + C =$

(a) $-\frac{7}{4}$

(b) -1

(c) $\frac{3}{2}$

(d) $\frac{5}{7}$

(e) 0

8. The differential of $y = e^{\cot(\pi x)}$ is

(a) $dy = (-\pi \csc^2(\pi x) e^{\cot(\pi x)})dx$

(b) $dy = (\sec^2(\pi x) e^{\cot(\pi x)})dx$

(c) $dy = (-\pi \sec(\pi x) \tan(\pi x) e^{\cot(\pi x)})dx$

(d) $dy = e^{\cot(\pi x)}dx$

(e) $dy = \cot(\pi x)dx$

9. The graph of $f(x) = \dfrac{x^2 - 5x + 6}{x^2 - 6x + 8}$ has

(a) one horizontal asymptote and one vertical asymptote

(b) one horizontal asymptote and two vertical asymptotes

(c) one slant asymptote and one vertical asymptote

(d) two horizontal asymptotes & two vertical asymptotes

(e) one slant asymptote and one horizontal asymptote

10. The polynomial $f(x) = 1 + 2x + 6x^2 - x^4$ is [CD: concave downward, CU: concave upward]

(a) CD on $(-\infty, -1)$ and $(1, \infty)$; CU on $(-1, 1)$

(b) CD on $(-\infty, 1)$; CU on $(1, \infty)$

(c) CD on $(-\infty, 2)$ and $(5, \infty)$; CU on $(2, \infty)$

(d) CD on $(-\infty, \infty)$

(e) CD on $(-3, 3)$; CU on $(-\infty, -3)$ and $(3, \infty)$

11. The **sum** of the critical numbrs of $f(x) = (x-1)^{3/5} \cdot (4-x)$ is

(a) $\frac{25}{8}$

(b) $\frac{17}{8}$

(c) $-\frac{9}{8}$

(d) 0

(e) $\frac{3}{7}$

12. If M and m are respectively the absolute maximum and absolute minimum values of $f(x) = \cos x + \sin x$ on $[0, \pi]$, then $\sqrt{2}\,M + m =$

(a) 1

(b) $2\sqrt{2}$

(c) 3

(d) 0

(e) -1

13. If $2x^2 + 3y^2 = 18$, then $y^3 y'' =$

(a) -4

(b) $\frac{8}{9}$

(c) $-\frac{1}{6}$

(d) 3

(e) -6

14. $\lim_{x \to 0} \frac{x - \tan x}{x - \sin x} =$

(a) -2

(b) 2

(c) 1

(d) -1

(e) 0

15. Using differentials (or a linear approximation), the value of $(64.018)^{2/3}$ is approximately equal to

(a) 16.003

(b) 16.01

(c) 4.003

(d) 16.018

(e) 12.002

16. Using Newton's Method to approximate one root of the equation $(x-2)^4 = \ln x$, we find that if $x_1 = 1$, then $x_2 =$

(a) $\frac{6}{5}$

(b) $\frac{4}{3}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

(e) $\frac{3}{5}$

17. If $f'(t) = 2t - 3\sin t$ and $f(0) = 5$, then $f(\pi) =$

(a) $\pi^2 - 1$

(b) $\pi^2 + 5$

(c) $\pi - 3$

(d) $\pi^2 + 2$

(e) 2π

18. Let a be a positive real number such that $a \neq e$. The **slope of the tangent line** to the curve $x^y = y^x$ at the point (a, a) is equal to

(a) 1

(b) $a \ln a$

(c) $\ln a - 1$

(d) $-a$

(e) a^2

19. The function $f(x) = \ln(x^2 - 3x + 2)$ has

(a) neither local minimum nor local maximum

(b) one local maximum and no local minimum

(c) one local maximum and two local minima

(d) one local minimum and two local maxima

(e) one local minimum and one local maximum

20. If $\sinh x + \cosh x = 5$, then $\tanh x =$

(a) $\frac{12}{13}$

(b) $\frac{1}{5}$

(c) $\frac{21}{25}$

(d) $\frac{3}{4}$

(e) $\frac{13}{16}$

21. $\dfrac{d}{dt}[\ln(\cosh t) - \dfrac{1}{2}\tanh^2 t] =$

(a) $\tanh^3 t$

(b) $\tanh t - \operatorname{sech}^2 t$

(c) 0

(d) $\coth t + \tanh t \operatorname{sech} t$

(e) $\tanh t - \tanh^2 t$

22. The function $f(x) = \dfrac{\sqrt{1-x^2}}{x}$ is

(a) decreasing on $(-1, 0)$ and on $(0, 1)$

(b) increasing on $(-1, 0)$ and on $(0, 1)$

(c) increasing on $(-1, 0)$ and decreasing on $(0, 1)$

(d) increasing on $(0, 1)$ and dcreasing on $(-1, 0)$

(e) decreasing on $(-\infty, \infty)$

23. $\lim_{x\to 0^+} (1-4\sin(3x))^{5\cot(9x)} =$

(a) $e^{-20/3}$

(b) $e^{-5/4}$

(c) e^{-20}

(d) $e^{-15/4}$

(e) 1

24. The equation of **the normal line** to the curve

$$y = x^4 - 10x + 11$$

that is parallel to the line $x - 6y = 3$ is

(a) $y = \frac{1}{6}x + \frac{11}{6}$

(b) $y = \frac{1}{6}x - \frac{5}{3}$

(c) $y = -6x - \frac{11}{6}$

(d) $y = -6x + 3$

(e) $y = -\frac{1}{6}x + \frac{1}{3}$

25. Two sides of a triangle are 5 m and 8 m in length and the angle between them is increasing at a rate of 0.3 rad/s. The rate at which **the area of the triangle** is increasing when the angle between the sides of fixed length is $\frac{\pi}{3}$ is

(a) 3 m^2/s

(b) 10 m^2/s

(c) 2.5 m^2/s

(d) 6.5 m^2/s

(e) 5 m^2/s

26. The sum of two positive numbers is 5. If the product P of the square of the first number and the cube of the second number is **maximized**, then $P =$

(a) 108

(b) 25

(c) 16

(d) 64

(e) 72

27. If $f(t) = 5 + 6\sin(3t)$, then $f^{(21)}\left(\frac{\pi}{3}\right) =$

(a) $-6 \cdot 3^{21}$

(b) 3^{21}

(c) 0

(d) $6 \cdot 3^{21}$

(e) $(21)!$

28. Applying the Mean Value Theorem to $f(x) = \tan^{-1} x$ on the interval $[1, 2]$, we conclude that

(a) $\frac{\pi}{4} + \frac{1}{5} < \tan^{-1} 2 < \frac{\pi}{4} + \frac{1}{2}$

(b) $\frac{\pi}{2} < \tan^{-1} 2 < \pi$

(c) $\frac{\pi}{4} + \frac{1}{2} < \tan^{-1} 2 < \frac{\pi}{2}$

(d) $\frac{1}{5} < \tan^{-1} 2 < \frac{1}{2}$

(e) $\frac{\pi}{8} < \tan^{-1} 2 < \frac{\pi}{4} + \frac{1}{5}$